

Facebook (Meta Platforms)

A deep dive into the world's largest social network — its features, architecture, and the technology powering billions of connections every day.



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Introduction

What is Facebook?

The Company

Facebook is the flagship social networking platform of **Meta Platforms, Inc.** — one of the world's most valuable technology companies. Originally launched as a college networking site, it has grown into a global communication and media giant.

- **Founded:** February 4, 2004
- **Founder:** Mark Zuckerberg (along with Eduardo Saverin, Andrew McCollum, Dustin Moskovitz, and Chris Hughes)
- **Headquarters:** Menlo Park, California, USA
- **Rebranded:** Meta Platforms, Inc. in October 2021

Mission

"Give people the power to build community and bring the world closer together."

Scale

3B+

Monthly Active Users

2004

Year Founded

Key Features

Facebook's core product is built around a rich set of social interaction tools that keep users engaged and connected.



Posts

Share text, photos, videos, and links with friends, groups, or the public. Posts are the primary unit of content on Facebook.



Likes & Reactions

Express emotions on any post with Like, Love, Haha, Wow, Sad, or Angry reactions — a simple but powerful engagement signal.



Comments

Engage in threaded conversations directly on posts, photos, and videos. Comments support text, emojis, GIFs, and images.



Messaging

Facebook Messenger enables real-time private and group messaging, voice calls, and video calls across devices.



News Feed

A personalized, algorithmically curated stream of posts, stories, ads, and recommendations from friends and pages you follow.

Why Facebook is Powerful

Billions of Users

With over **3 billion monthly active users**, Facebook is the single largest social platform on Earth. Its network effect means every new user makes the platform more valuable for everyone else. No other social network has achieved this scale of global reach across so many demographics and geographies.

Real-Time Communication

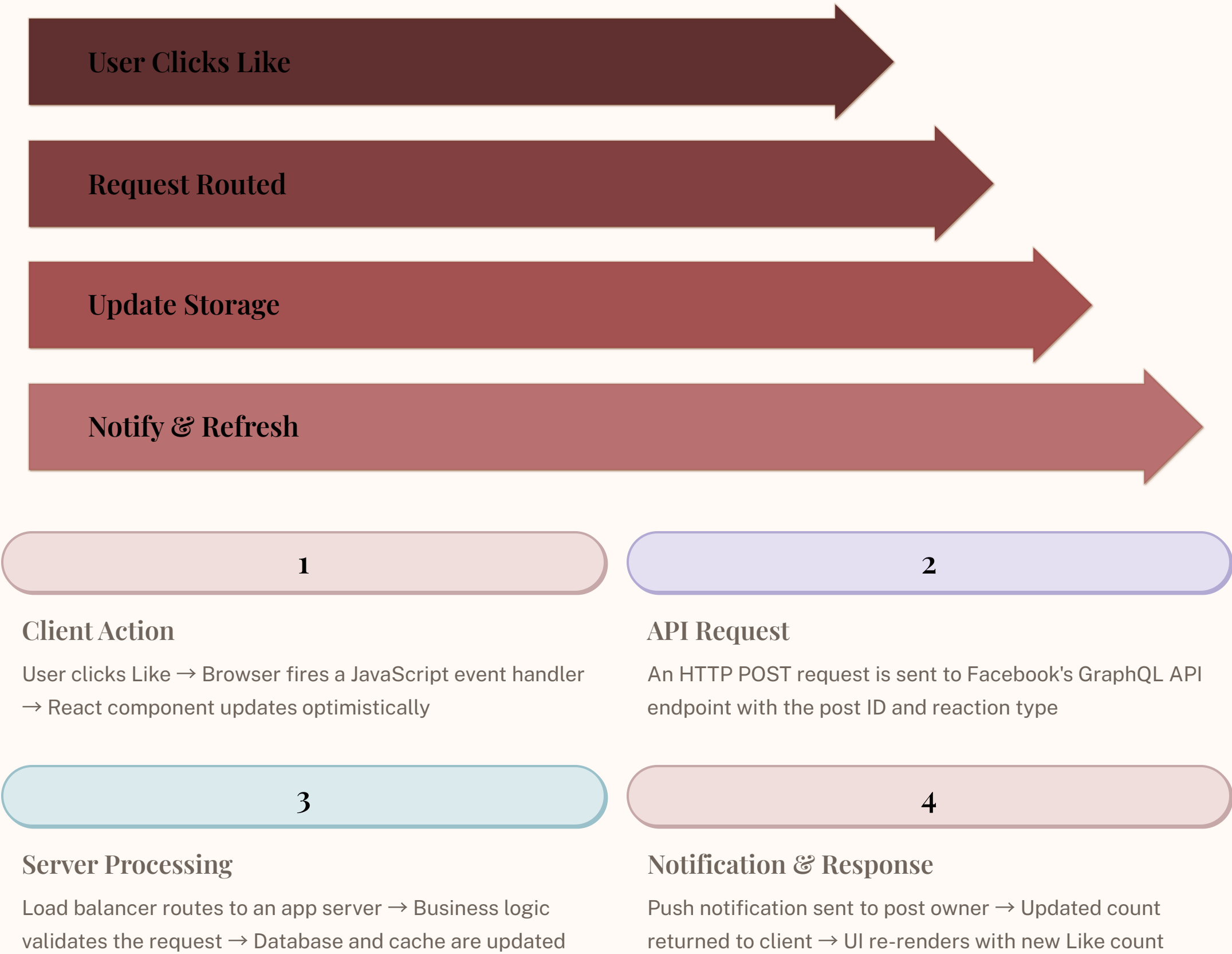
Facebook enables **real-time communication** at a massive scale — from instant messaging and live video streaming to real-time notifications and group interactions. Events, crises, and conversations unfold live, making Facebook a primary source of breaking news and social coordination worldwide.

AI-Based Personalization

Facebook's **AI and machine learning systems** power everything from the News Feed ranking algorithm to content recommendations, ad targeting, facial recognition, and spam detection. The platform learns from every click, scroll, and interaction to deliver a uniquely personalized experience to each of its billions of users.

What Happens When a User Clicks "Like"

A seemingly simple action triggers a sophisticated chain of events across the client-server architecture.



Frontend Technologies

Facebook's user interface is built with a modern, component-driven frontend stack designed for performance and scale.



HTML

HyperText Markup Language forms the structural backbone of every Facebook page. Facebook uses semantic HTML5 to ensure accessibility and SEO compatibility across its web interfaces.



JavaScript

JavaScript powers all client-side interactivity on Facebook. Facebook also created **Flow** (a static type checker for JS) and **Jest** (a testing framework) — both open-source tools born from their JS needs.



CSS

Cascading Style Sheets handle all visual styling. Facebook uses a custom CSS-in-JS approach and atomic CSS patterns to manage styles at scale across thousands of UI components.



React

React was invented by Facebook engineer Jordan Walke and open-sourced in 2013. It powers Facebook's entire UI with a component-based, virtual DOM architecture enabling fast, declarative rendering at massive scale.



Facebook also created **GraphQL** — a query language for APIs — and **Relay**, a React framework for data fetching, both of which are central to their frontend architecture.

Backend Technologies

Facebook's backend is a polyglot environment — different languages are chosen for different tasks based on performance, safety, and scalability requirements.



PHP

Facebook was originally built entirely in **PHP**. To overcome PHP's performance limitations at scale, Facebook created **HipHop for PHP (HPHPc)**, a PHP-to-C++ transpiler, and later the **HHVM** (HipHop Virtual Machine) runtime.



Hack

Hack is a programming language developed by Facebook as a successor to PHP. It runs on HHVM and adds static typing, generics, and other modern language features while maintaining PHP compatibility for rapid development.



Python

Python is used extensively at Facebook for infrastructure tooling, automation scripts, data pipelines, and machine learning workflows. Its rich ecosystem of ML libraries (PyTorch, NumPy) makes it indispensable for AI research and production.



C++

C++ is used for performance-critical systems at Facebook, including the **Thrift** RPC framework, parts of the storage layer, and low-latency services where raw speed is essential. Facebook's Folly library is a widely used C++ utility library.



Java

Java powers several of Facebook's large-scale distributed systems and Android mobile development. It is used in data processing pipelines, backend services, and tooling that benefits from the JVM ecosystem and strong typing.

Database Systems

Facebook manages petabytes of data across billions of users using a combination of relational, NoSQL, and distributed storage systems.

MySQL

Facebook is one of the world's largest users of **MySQL**, the open-source relational database. They store core user data, profiles, and social graph relationships in MySQL. Facebook has contributed heavily to MySQL's development and built **MyRocks** — a MySQL storage engine based on RocksDB — to improve storage efficiency.

Cassandra

Apache Cassandra was originally developed at Facebook and later open-sourced. It is a distributed NoSQL database designed for high availability and linear scalability with no single point of failure. Facebook uses Cassandra for high-write-throughput workloads like inbox search and activity feeds.

Hadoop

Apache Hadoop forms the backbone of Facebook's data warehouse infrastructure. Facebook runs one of the world's largest Hadoop clusters, processing **petabytes of data daily** for analytics, reporting, machine learning training, and business intelligence through their internal data platform called **Hive** (also created at Facebook).

  Facebook also uses **TAO** (The Associations and Objects) — a custom distributed data store built on top of MySQL — specifically optimized for reading and writing social graph data at scale.

Other Services

Beyond the core stack, Facebook relies on a suite of specialized infrastructure services to deliver a fast, reliable, and intelligent experience at global scale.

CDN (Content Delivery Network)

Facebook uses a massive global **CDN** to serve static assets — images, videos, JavaScript bundles, and CSS — from edge servers close to the user. This dramatically reduces latency and load times worldwide. Facebook also operates its own CDN infrastructure called **Facebook Points of Presence (PoPs)**.

AI / ML

Facebook's **AI Research (FAIR)** lab and applied ML teams power News Feed ranking, content moderation, ad targeting, facial recognition (DeepFace), translation, and recommendation systems. **PyTorch**, Facebook's open-source deep learning framework, is the foundation of most ML workloads.

Caching

Facebook uses **Memcached** at an extraordinary scale — their deployment is one of the largest in the world. Caching sits in front of the database layer to absorb billions of read requests per second, reducing database load and ensuring sub-millisecond response times for frequently accessed data like user profiles and Like counts.

Real-Time Messaging

Facebook Messenger uses a combination of **MQTT** (a lightweight publish-subscribe protocol), **WebSockets**, and custom push notification infrastructure to deliver real-time messages with minimal latency. The system handles billions of messages per day across mobile and web clients simultaneously.

Conclusion

Facebook (Meta Platforms) is far more than a social network — it is a **technological marvel** built on decades of engineering innovation, open-source contributions, and relentless scaling.



Unmatched Scale

With **billions of users**, real-time communication, and AI-based personalization, Facebook remains the world's most powerful social platform.



Full-Stack Innovation

From **React and GraphQL** on the frontend to **Hack, C++, and Python** on the backend, Facebook has invented or popularized many of the technologies the modern web runs on.



Data at Petabyte Scale

MySQL, Cassandra, Hadoop, and TAO work in concert to store, retrieve, and analyze petabytes of social data reliably and efficiently every single day.



Infrastructure Excellence

A global **CDN, Memcached caching layer, AI/ML pipelines, and real-time messaging** infrastructure ensure that every Like, comment, and message is delivered instantly and reliably to users around the world.



Make slides visually appealing with diagrams, icons, and minimal text. Use a modern tech theme.